EDO STATE COLLEGE OF AGRICULTURE AND NATURAL RESURCES, IGUORIAKHI, NIGERIA

DEPARTMENT OF AGRICULTURAL TECHNOLOGY

COURSE CODE: AGT 101

COURSE TITLE: PRINCIPLES OF CROP PRODUCTION

CREDIT LOAD: 3 UNITS

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GOAL: To acquaint the students with the principles and practices of crop production.

General Objectives: On completion of this course the student should be able to:

1.0 Identify and understand the scope of crop production in Nigeria.

2.0 Understand the different cropping systems.

3.0 Understand the principles and practices of tillage.

4.0 Understand the different methods of propagating plants.

5.0 Understand and practice successful weed control in crop production

6.0 Understand the principles and practices of manuring and fertilising.

7.0 Understand the principles and practices of crop protection.

8.0 Understand the principles and practices of harvesting, storage and product handling.

UNIT 1: CONCEPTS OF CROP PRODUCTION

Introduction

Firstly, we must understand that a **crop** is a plant which is cultivated by man for some beneficial purposes. It is worthy to note that all crops are plants but not all plants are crops.

Crop production is the process of cultivating plants. It is a branch of agriculture that involves the selection, preparation, and planting of seeds, as well as the care and management of crops until they are harvested.

Objectives of crop production

1. To produce food and other agricultural products to meet the needs of the population. *This is the primary objective of crop production, and it is essential for ensuring food security and nutrition for all.*
2. To maximize crop yields while minimizing environmental impact. *Crop producers strive to produce as much food as possible while using resources efficiently and minimizing their impact on the environment. This can be achieved through sustainable farming practices, such as crop rotation, cover cropping, and integrated pest management.*
3. To produce crops that are high in quality and nutritious. *Crop producers aim to produce crops that are safe to eat and provide the nutrients that people need to stay healthy. This can be achieved through careful crop selection, management, and postharvest handling.*
4. To produce crops that are resistant to pests and diseases. *Crop producers want to produce crops that are not easily damaged by pests and diseases. This can be*

achieved through crop breeding, crop rotation, and other integrated pest management practices.

5. To produce crops that are adapted to the local climate and soil conditions. *Crop producers need to select crops that can grow well in the local climate and soil conditions. This is important for maximizing crop yields and reducing the risk of crop failure.*
6. To generate income for farmers and other people involved in the agricultural sector. *Crop production is a vital part of the global economy and provides jobs and income for millions of people.*

Explanations…

<u>Importance of crop production to agriculture</u>

- It provides food for the population. Crop production is the primary source of food for humans and animals. Crops such as cereals, legumes, fruits, and vegetables provide us with the essential nutrients we need to survive and thrive.
- It provides fiber for clothing and other products. Crops such as cotton, flax, and hemp are used to produce fiber for clothing, textiles, and other products.
- It provides fuel for transportation and other purposes. Crops such as sugarcane, corn, and soybeans are used to produce biofuels such as ethanol and biodiesel.
- It provides medicinal products. Crops such as herbs and spices have been used for centuries to treat a variety of ailments.
- It provides income for farmers and other people involved in the agricultural sector. Crop production is a major economic driver in many countries around the world and provides jobs and income for millions of people.

<u>Scope of crop production in Nigeria</u>

Nigeria is a major agricultural producer, and crop production remains the largest driver of the Agricultural sector. The scope of crop production in Nigeria is vast and includes a wide range of activities, such as:

- **Seed production:** The production of high-quality seed is essential for improving crop yields. Nigeria has a number of seed production companies that produce seed for a variety of crops, including cereals, legumes, vegetables, and fruits.
- **Land preparation:** Land preparation involves clearing the land of debris, weeds, and pests. It also involves tilling the soil to improve drainage and aeration.
- **Planting:** Planting involves sowing seeds or transplanting seedlings into the ground. The planting method will vary depending on the type of crop being planted.
- **Crop management:** Crop management includes irrigation, fertilization, weed control, and pest and disease control. Crop producers use a variety of methods to manage their crops, including manual labour, machinery, and chemicals.

- **Harvesting:** Harvesting involves collecting the crops from the field. The harvesting method will vary depending on the type of crop being harvested.
- **Post-harvest handling:** Post-harvest handling includes drying, storing, and transporting the crops. Post-harvest handling is important to ensure that the crops are of high quality and can be stored for long period of time.

The main crops grown in Nigeria include:

- Food crops: Cereals (rice, maize, millet, sorghum), tubers (cassava, yams, potatoes), legumes (beans, cowpeas, peanuts), vegetables, and fruits.
- Cash crops: Cocoa, rubber, oil palm, cotton, and groundnuts.

Nigeria is a self-sufficient producer of most food crops, and it also exports some cash crops, such as cocoa and rubber. However, the country still imports some food crops, such as wheat and rice.

Principles and application of crop production in agriculture

The principles of crop production are the basic concepts that underlie the cultivation of plants. These principles are based on our understanding of plant biology, ecology, and the environment.

Some of the key principles of crop production include:

- **Plant nutrition:** Crops need a variety of nutrients to grow and develop properly. These nutrients can be obtained from the soil, water, and air. Crop producers use a variety of methods to ensure that their crops have the nutrients they need, such as fertilization and crop rotation.
- **Water management:** Crops need water to grow and develop. Crop producers must manage the water supply carefully to ensure that their crops have enough water to meet their needs. This may involve irrigation, drainage, and water conservation practices.
- **Pest and disease management:** Pests and diseases can damage crops and reduce yields. Crop producers use a variety of methods to control pests and diseases, such as crop rotation, pesticide use, and biological control.
- **Soil management:** Soil is the foundation of crop production. Crop producers must manage the soil carefully to maintain its fertility and structure. This may involve tillage practices, crop rotation, and the use of cover crops.
- **Climate adaptation:** Crops are affected by the climate in which they are grown. Crop producers must select crops that are adapted to the local climate and take steps to mitigate the effects of climate change.

Application of crop production principles

The principles of crop production can be applied in a variety of ways to improve crop yields and reduce environmental impact. Some examples include:

- Crop rotation: Crop rotation is the practice of growing different crops in the same field in a sequence. Crop rotation helps to improve soil fertility, reduce pests and diseases, and conserve water.
- Cover cropping: Cover cropping is the practice of growing a crop of plants to cover the soil between rows of crops or during the off-season. Cover crops can help to improve soil health, reduce erosion, and suppress weeds.
- Integrated pest management (IPM): IPM is a holistic approach to pest management that emphasizes prevention and the use of biological controls. IPM can help to reduce the use of pesticides and protect the environment.
- Precision agriculture: Precision agriculture is a farming approach that uses technology to monitor and manage crops more precisely. Precision agriculture can help to reduce the use of inputs, such as fertilizer and pesticides, while increasing crop yields.

Factors affecting crop production

Crop production is affected by a wide range of factors, including environmental factors, economic factors, and social factors.

Environmental factors

Environmental factors are the physical and biological conditions in which crops are grown. Some of the key environmental factors that affect crop production include:

- **Climate:** Climate is a major factor affecting crop production. Crops need a certain amount of sunlight, water, and temperature to grow and develop. Climate change is a major threat to crop production, as it is leading to more extreme weather events such as droughts, floods, and heat waves.
- **Soil:** Soil quality is also important for crop production. Crops need nutrient-rich soil that is well-drained. Land degradation, such as deforestation, overgrazing, and soil erosion, can reduce soil quality and make it more difficult to grow crops.
- **Pests and diseases:** Pests and diseases can damage crops and reduce yields. Crop producers must use a variety of methods to control pests and diseases, such as crop rotation, pesticide use, and biological control.
- **Weeds:** Weeds can compete with crops for water, nutrients, and sunlight. Crop producers use a variety of methods to control weeds, such as herbicides, tillage, and hand-weeding.

Economic factors

Economic factors also affect crop production. Some of the key economic factors that affect crop production include:

- **Input costs:** The cost of inputs such as seed, fertilizer, and pesticides can significantly affect crop production costs. Crop producers must carefully manage their input costs to ensure that their crops are profitable.
- **Output prices:** The prices of agricultural products can fluctuate significantly, which can affect the profitability of crop production. Crop producers must carefully consider market conditions when making decisions about crop selection and production.
- **Government policies:** Government policies, such as agricultural subsidies and trade restrictions, can also affect crop production. Crop producers must be aware of government policies and how they may affect their business.

Social factors

Social factors also affect crop production. Some of the key social factors that affect crop production include:

- **Labour availability:** The availability of labor can affect crop production, especially in labor-intensive crops. Crop producers must carefully manage their labor resources to ensure that their crops are harvested and marketed in a timely manner.
- **Land tenure:** Land tenure, or the system of land ownership and use, can also affect crop production. Crop producers who own their land are more likely to invest in long-term improvements, such as soil conservation and irrigation systems.
- **Access to credit:** Access to credit can help crop producers to purchase inputs, invest in new technologies, and expand their operations. Crop producers without access to credit may be limited in their ability to improve their production practices and increase their yields.

How to mitigate the impact of negative factors

There are a number of things that crop producers can do to mitigate the impact of negative factors on crop production. Some examples include:

- Diversify crops: Diversifying crops can help to reduce the risk of crop failure due to pests, diseases, or weather events.
- Use sustainable farming practices: Sustainable farming practices can help to improve soil health, reduce erosion, and enhance biodiversity. This can lead to higher crop yields and reduced environmental impact.
- Invest in new technologies: New technologies such as precision agriculture and drought-resistant crop varieties can help crop producers to improve crop yields and reduce their vulnerability to environmental factors.

- Advocate for favorable government policies: Crop producers can advocate for government policies that support sustainable agriculture and provide financial assistance to farmers.

UNIT 2: CROPPING SYSTEMS

Cropping system is defined as the order in which crops are cultivated on a piece of land over a fixed period. Cropping system is the way in which different crops are grown. Also**,** the term cropping systems refers to the crops and crop sequences and the management techniques used on a particular field over a period of years. The objective of any cropping system is the efficient utilization of all resources such as land, water and solar radiation, maintaining stability in production and obtaining higher net returns.

Cropping systems are classified based on the following criteria

1. The distribution of crops in time, i.e. whether shifting cultivation, continuous cropping, monoculture, or crop rotation is practiced.
2. The distribution of the crops in space on the field, i.e. whether intercropping or sole cropping is practiced.
3. The level of management and resources utilized to produce the crop, i.e. whether production is intensive or extensive.
4. The type of crop grown, i.e. whether orchard, arable cropping, pasturing, forestry, etc. is practiced.

A. Shifting Cultivation

In this system, the farm is not at a permanent location. Instead, a piece of land is cleared, farmed for a few years and then abandoned in preference for a new site. While the new site is being farmed, natural vegetation is allowed to grow on the old site. Eventually, after several years of bush fallows, the farmer returns to the original location. The practice of moving the home along with the farm is discontinued and in its place the practice of making home stationary is common in tropical Africa.

Advantages and disadvantages of shifting cultivation

The advantages include keeping the soil sufficiently fertile when there is abundant available land for farming and preventing the spread of insect pests, other pests and plant pathogens. The disadvantages are greater, and include inadequate cultivable land for food production, requirement of large land area, inadequate time for soil fertility restoration and waste of farmers' energy resources in frequent slashing of agricultural fields.

B. Bush Fallow

In bush fallow systems, a piece of land is cultivated for a period of time and then left uncultivated for several years so that its fertility will be restored. During the resting period fertility is restored to an extent depending on the length of fallow, nature of vegetation and the soil. It involves fixed settlements, but periodic shifting rotation of fields within the cultivated land. In the early times, fallow fields may be left untilled or tilled but not planted for the fallow period. Sometimes, the fallow fields were used for pasturage for animals, which had the incidental benefit of fertilizing the soil. Fallow types may be natural (colonized by invading weeds), or planted with quick, improved fertility-regenerating leguminous (weed) species e.g. *Chromoleana ordorata, Mucuna utilis, Centrosema pubescens, Crotalaria spp*., etc. Planted green fallows are common in some parts of Southern Nigeria in response to high population pressure on available land. An added advantage of this is the superior competitive ability of the green fallow species over the native weeds in the fallow, although the practice requires huge time and money investments in seed sowing, which may reduce the cost/benefit ratio.

C. <u>Continuous Cropping</u>

In contrast to shifting cultivation, continuous cropping implies the cultivation of the same piece of land year after year. Fallowing may occur, but it never occurs more than a season or two. The absence of a protracted fallow periods means that other soil management practices must be employed in order to maintain high soil fertility.

Agricultural practices for maintaining soil fertility under continuous cropping

1. Application of fertilizers and other soil amendments in order to boost fertility.
2. Judicious selection of the crops and crop combinations to be grown. Crop rotations and carefully planned intercrop combination are indispensable.
3. Introducing short term fallow periods in to the cropping cycle. A leguminous cover crop can be planted on the fallow land so as to aid the fixation of nitrogen by legumes during the fallow period and through increasing the soil organic matter content when the fallow crop is ploughed under

Advantages of Continuous Cropping

1. Land utilization under continuous cropping is extremely efficient. A very high percentage of land is under crops at any given time.
2. It is possible and economically feasible, to erect permanent structures on the farm site.

D. <u>Crop Rotation</u>

The practice of growing different kinds of crops, one at a time, in a definite sequence on the same piece of land is referred to as crop rotation. In designing a good crop rotation, the farmer must decide what crops to have in the rotation, in what sequence the crops should occur, and for how many years or season each cycle of the rotation must run. A good rotation that provides for maintenance or improvement of soil productivity usually includes a legume crop to promote fixation of nitrogen, a grass or legume sod crop for maintenance of humus, a cultivated crop for weed control and fertilizers. Perennial legumes and grasses may leave two to three tons of dry weight per acre of roots residues in the soil when plowed down.

Factors to consider in deciding the sequence of crops (principles of crop rotation)

1. The target crop (the main crop) should be planted immediately after the legumes or fallow period. At this time the fertility of the soil is at its peak and the optimum realizable yield of the target crop is possible. Crops which are known to have a high demand for nutrients are also timed for the first season after the fallow.
2. Crops which are deep feeders should alternate with shallow feeders. In this way, nutrient removal occurs uniformly from the various soil layers rather than occurring in only one layer.
3. Crops that are botanically similar or are likely to be attacked by the same diseases and pests should not normally follow each other in the rotation. Yams, for example, should not follow cowpeas in rotation if the root-knot nematode is prevalent, as the nematodes left over from the cowpea crop will severely reduce yam yields. However, if the nematode problem does not exist in the area, yam could conveniently follow cowpeas.
4. The number of years for which each cycle of the rotation should run is determined by the number of crops in the rotation, the length of their growing seasons and how frequent the farmer can grow the target crop without running into problems of disease and soil fertility. For example, the time interval between the harvesting of the target crop and its being planted again on the same piece of land should be long enough to prevent the carryover of pathogens in crop residues from one cycle to the next.

Examples of crop rotation

A simple example of crop rotation would be to grow a cereal crop, such as wheat, followed by a legume crop, such as beans. The cereal crop would deplete the soil of nitrogen, while the legume crop would fix nitrogen from the atmosphere. This would improve the soil fertility and make it more productive for the next crop.

	Year 1	Year 2	Year 3
Plot 1	Cereal	Legume	Cereal
Plot 2	Legume	Cereal	Legume

Another example of crop rotation would be to grow a deep-rooted crop, such as carrots, followed by a shallow-rooted crop, such as lettuce. The deep-rooted crop would improve drainage and aeration in the soil, while the shallow-rooted crop would benefit from the improved soil conditions.

	Year 1	Year 2	Year 3
Plot 1	Deep-rooted	Shallow-rooted	Deep-rooted
Plot 2	Shallow-rooted	Deep-rooted	Shallow-rooted

Advantages of crop rotation

- Improved soil health: Crop rotation helps to improve soil health by increasing soil organic matter, improving drainage and aeration, and reducing soil erosion.
- Reduced pest and disease pressure: Crop rotation helps to reduce pest and disease pressure by disrupting the life cycles of pests and diseases, and reducing the buildup of pest populations.
- Increased crop yields: Crop rotation can help to increase crop yields by improving soil fertility, reducing competition between crops, and reducing pest and disease damage

Disadvantages of crop rotation

- Reduced flexibility: Crop rotation can reduce flexibility for farmers, as they must plan their crops in advance.
- Increased labour and equipment requirements: Crop rotation can increase labor and equipment requirements, as farmers may need to plant different crops at different times of the year and use different equipment for different crops.
- Reduced income in the short term: Crop rotation may reduce income in the short term, as farmers may need to plant less profitable crops in order to improve soil health and reduce pest and disease pressure.
- The risk of crop failure is ever present, and the risk is greater where only one crop is grown.

E. <u>Monoculture</u>

Monoculture is the practice of growing a single crop on a piece of land year after year. It is a common agricultural practice that is used to produce large quantities of a particular crop.

Advantages of monoculture

- Increased efficiency: Monoculture can be more efficient than other farming practices, as farmers can specialize in growing a single crop. This can lead to lower production costs and higher yields.
- Simplified management: Monoculture can simplify crop management, as farmers only need to learn about the needs of a single crop.

Disadvantages of monoculture

- Depleted soil fertility: Monoculture can deplete soil fertility, as the same crop is constantly drawing nutrients from the soil. This can lead to lower crop yields over time.
- Increased pest and disease pressure: Monoculture can increase pest and disease pressure, as pests and diseases can easily develop resistance to crops that are grown year after year.

- Reduced biodiversity: Monoculture reduces biodiversity, as only a single crop is grown in a given area. This can have negative consequences for the environment, such as increased soil erosion and reduced water quality.

F. <u>Intercropping</u>

The practice of growing one crop variety in pure stands on a field is referred to **as sole cropping.** In this practice, only one crop variety occupies the land at any one time. The alternative practice of growing two or more crops simultaneously on the same field is called **intercropping.**

Intercropping is the practice of growing two or more crops on the same piece of land at the same time in a spatial arrangement. Intercropping is a traditional agricultural practice that is becoming increasingly popular in modern agriculture.

Types of intercropping

- Row intercropping: Row intercropping is the simplest type of intercropping. It involves planting two or more crops in alternating rows.
- Strip intercropping: Strip intercropping is similar to row intercropping, but the crops are planted in wider strips.
- Mixed intercropping: Mixed intercropping is the most complex type of intercropping. It involves planting two or more crops randomly throughout the field.
- Relay inter-cropping: This is when a second crop variety is sown between the stands of an existing sole crop just before the first crop is harvested. As such, both the first and second crops spend most of their field lives as sole crop, and grow together on the field for only a brief period.

Advantages of intercropping

- Increased crop yields: Intercropping can increase crop yields by reducing competition between crops for resources such as water, light and nutrients, increasing utilization of land and other resources, and promoting beneficial interactions between crops.
- Reduced pest and disease pressure: Intercropping can reduce pest and disease pressure by disrupting the life cycles of pests and diseases, reducing the buildup of pest populations, providing a habitat for beneficial insects.
- Improved soil health: Intercropping can improve soil health by increasing soil organic matter, improving drainage and aeration, and reducing soil erosion.

Disadvantages of intercropping

- Increased labour and equipment requirements: Intercropping can increase labor and equipment requirements, as farmers may need to plant different crops at different times of the year and use different equipment for different crops.
- Since many crops exist together on the field, it is not possible to tailor production practices to the needs of any particular crop.
- Control of pests and diseases is particularly difficult because pesticides which have been developed to control a disease on one particular component crop may have deleterious effect on other crops in the combination.
- Increased risk of crop failure: Intercropping can increase the risk of crop failure, as if one crop fails it can affect the other crops.

Key factors influencing intercropping systems

1. Maturity of crops
2. Selecting compatible crop/ choice of crop species
3. Plant population
4. Plant spacing
5. Time of planting

UNIT 3: UNDERSTAND THE DIFFERENT METHODS OF PROPAGATING PLANT

The success of fruit orchards and vegetable farms depends on reliable planting material. Diseased and genetically inferior plants have catastrophic effects on the productivity of fruit industry. All plants multiply themselves by sexual propagation (seed) or by asexual propagation (vegetative) methods. However, researchers/propagators have developed some other techniques for the rapid and better multiplication of plants. Various plant species occupy a wide variety of habitats. To be able to adapt to the environment, they have to device method to perpetuate in the particular environment by producing their offspring to survive. A vegetative reproduction is the process of multiplication in which a portion of fragment of the plant body functions as propagules and develops into a new individual plant which involves the production of new plants without the act of fertilization or sexual union. Further can be said that, vegetative propagation of the plant is a form of plant propagation in which the new individual plant arises from any vegetative part of the parents (root, stem, leaf and other organs), and possesses the same characteristics of their parent plant from which it was obtained.

Vegetative propagation (asexual propagation)

In higher plants, any part of the body may be capable of vegetative propagation. Many plants produce modified stems, roots, and leaves, especially for natural vegetative propagation. In the other words, asexual means of reproduction produce new individuals without the fusion of gametes, genetically identical to the parent plants and each other, except when sudden change, 'mutation', occurs. Man has developed artificial vegetative propagation for many useful plants which are widely used in the horticultural industry. The main asexual methods of propagation include:

A. CUTTINGS

Many types of plants, both woody and herbaceous, are frequently propagated by cuttings. A cutting is a vegetative plant part that is severed from the parent plant to regenerate itself, thereby forming a whole new plant. Below are the various types of cuttings.

Stem Cuttings

Numerous plant species are propagated by stem cuttings. Some can be taken at any time of the year, but stem cuttings of many woody plants must be taken in the rainy season.

Leaf Cuttings

Leaf cuttings are used almost exclusively for a few indoor plants. The leaves of most plants will either produce a few roots but no plant or just decay.

Root Cuttings

Root cuttings are usually taken from 2 to 3-year-old plants during the wet season when they have a large carbohydrate supply. Root cuttings of some species produce new shoots, which then form their root systems, while root cuttings of other plants develop root systems before producing new shoots.

Layering

Stems still attached to their parent plants may form roots where they touch a rooting medium. Severed from the parent plant, the rooted stem becomes a new plant. This method of vegetative propagation, called layering, promotes a high success rate because it prevents the water stress and carbohydrate shortage those plague cuttings. Some plants layer themselves naturally, but sometimes plant propagators assist the process. Layering is enhanced by wounding one side of the stem or by bending it very sharply. The rooting medium should always provide aeration and a constant supply of moisture.

B. GRAFTING

Grafting and budding are methods of asexual plant propagation that join plant parts so they will grow as one plant. These techniques are used to propagate cultivars that will not root well as cuttings or whose own root systems are inadequate. One or more new cultivars can be added to existing fruit and nut trees by grafting or budding. The portion of the cultivar that is to be propagated is called the scion. It consists of a piece of the shoot with dormant buds that will produce the stem and branches. The rootstock, or stock, provides the new plant's root system and sometimes the lower part of the stem. The cambium is a layer of cells located between the wood and bark of a stem from which new bark and wood cells originate.

Four conditions must be met for grafting to be successful: 1. The scion and rootstock must be compatible 2. Each must be at the proper physiological stage 3. The cambial layers of the scion and stock must meet 4. The graft union must be kept moist until the wound has healed.

TYPES OF GRAFTING

Cleft grafting

Cleft grafting is often used to change the cultivar or top growth of a shoot or a young tree (usually a seedling). It is especially successful if done early into the wet season. Collect scion wood 3/8 to 5/8 inch in diameter. Cut the limb or small tree trunk to be reworked, perpendicular to its length. Make a 2-inch vertical cut through the center of the previous cut. Be careful not to tear the bark. Keep this cut wedged apart. Cut the lower end of each scion piece into a wedge. Prepare two scion pieces 3 to 4 inches long. Insert the scions at the outer edges of the cut in the stock. Tilt the top of the scion slightly outward and the bottom slightly inward to be sure the cambial layers of the scion and stock touch. Remove the wedge propping the slit open and cover all cut surfaces with grafting wax.

Bark grafting

Unlike most grafting methods, bark grafting can be used on large limbs, although these are often infected before the wound can completely heal. Collect scion wood 3/8 to 1/2 inch in diameter when the plant is dormant, and store the wood wrapped in moist paper in a plastic bag in the refrigerator. Saw off the limb or trunk of the rootstock at a right angle to itself. In the spring, when the bark is easy to separate from the wood, make a 12-inch diagonal cut on one side of the scion, and a 1½inch diagonal cut on the other side. Leave two buds above the longer cut. Cut through the bark of the stock, a little wider than the scion. Remove the top third of the bark from this cut. Insert the scion with the longer cut against the wood. Nail the graft in place with flat-headed wire nails. Cover all wounds with grafting wax.

Whip or tongue grafting

This method is often used for material 1/4 to ½ inch in diameter. The scion and rootstock are usually of the same diameter, but the scion may be narrower than the stock. This strong graft heals quickly and provides excellent cambial contact. Make one 2½-inch long sloping cut at the top of the rootstock and a matching cut on the bottom of the scion. On the cut surface, slice downward into the stock and up into the scion so the pieces will interlock. Fit the pieces together, then tie and wax the union.

Care of the Graft

Very little success in grafting will be obtained unless proper care is maintained for the following year or two. If a binding material such as a strong cord or nursery tape is used on the graft, this must be cut shortly after growth starts to prevent girdling. Rubber budding strips have some advantages over other materials. They expand with growth and usually do not need to be cut, as they deteriorate and break after a short time. It is also an excellent idea to inspect the grafts after 2 or 3 weeks to see if the wax has cracked, and if necessary, rewax the exposed areas. After this, the union will probably be strong enough and no more waxing will be necessary.

C. BUDDING

Budding, or bud grafting, is the union of one bud and a small piece of bark from the scion with a rootstock. It is especially useful when scion material is limited. It is also faster and forms a stronger union than grafting.

TYPES OF BUDDING

Patch budding

Plants with thick bark should be patch budded. This is done while the plants are actively growing, so their bark slips easily. Remove a rectangular piece of bark from the rootstock. Cover this wound with a bud and matching piece of bark from the scion. If the rootstock's bark is thicker than that of the scion, pare it down to meet the thinner bark so that when the union is wrapped the patch will be held firmly in place.

Chip budding

This budding method can be used when the bark is not slipping. Slice downward into the rootstock at a 45-degree angle through 1/4 of the wood. Make a second cut upward from the first cut, about one inch. Remove a bud and attending chip of bark and wood from the scion shaped so that it fits the rootstock wound. Fit the bud chip to the stock and wrap the union.

T-budding

This is the most commonly used budding technique. When the bark is slipping, make a vertical cut (same axis as the rootstock) through the bark of the rootstock, avoiding any buds on the stock. Make a horizontal cut at the top of the vertical cut (in a T shape) and loosen the bark by twisting the knife at the intersection. Remove a shield-shaped piece of the scion, including a bud, bark, and a thin section of wood. Push the shield under the loosened stock bark. Wrap the union, leaving the bud exposed.

Care of Buds

Place the bud in the stock during the rainy season. Force the bud to develop, then cut the stock off 5 to 6 cm above the bud. The new shoot may be tied to the resulting stub to prevent damage from the wind. After the shoot has made a strong union with the stock, cut the stub off close to the budded area.

OTHER TYPES OF ASEXUAL REPRODUCTION

Division

The division is another type of asexual reproduction that stands on its own. Here, plants with more than one rooted crown may be divided and the crowns planted separately. If the stems are not joined, gently pull the plants apart. If the crowns are united by horizontal stems, cut the stems and roots with a sharp knife to minimize injury. Divisions of some outdoor plants should be dusted with a fungicide before they are replanted. Examples: dahlias, iris, rhubarb, daylilies. Separation Separation is a term applied to a form of propagation by which plants that produce bulbs or corms multiply. For example • Bulbs New bulbs form beside the originally planted bulb. Separate these bulb clumps every 3 to 5 years for the largest blooms and to increase bulb population • Corms Large new corm forms on top of the old corm, and tiny cormels form around the large corm. After the leaves wither, dig up the corms and allow them to dry in indirect light for 2 or 3 weeks. Remove the cormels, and then gently separate the new corm from the old corm. Dust all-new corms with a fungicide and store them in a cool place until planting time.

Advantages of Asexual propagation

1. The horticultural crops which do not produce viable seeds are propagated by the vegetative method. 2. Most of the important fruit crops are cross-pollinated and are highly heterozygous. When propagated through seeds, the progenies show large variation, so vegetative propagation is a remedy for these crops. 3. The asexual propagation method gives true to type plants. 4. The vegetative way propagated plants bear fruits early. 5. In the case of fruit crops where rootstocks are used, the rootstocks impart insect or disease resistance to the plant. 6. Vegetative propagation helps to alter the size of the plant. i.e. dwarfing effect. This helps for spraying, intercropping & harvesting of crops easy and economical. 7. By grafting method different variety of fruit crop can be grown & harvested. 8. Inferior quality fruit plants can be converted into good quality plants. 9. Utilizing bridge grafting a repairing of injured plants can be done.

Disadvantages of the vegetative propagation

1. By vegetative propagation new variety cannot be developed. 2. It is an expensive method of propagation and required specialized skills. 3. The life span of vegetative propagated plants is short as compared to sexually propagated plants. 4. As all the plants are homozygous the whole plantation may get attacked by a particular pest or disease. 5. Viral diseases could be transferred through vegetative parts.

Sexual propagation

The Sexual reproduction method produces offspring by the fusion of gametes, resulting in offspring genetically different from the parent plants due to genetic exchange that occurred during fertilization which came from both parents. The outcome of this fusion is the seed. • A seed is an embryonic plant enclosed in a protective outer covering. The formation of the seed is part of the process of reproduction in seed plants, the spermatophytes, including the gymnosperm and angiosperm plants. • Seeds are the product of the ripened ovule, after fertilization by pollen and some growth within the mother plant. The embryo develops from the zygote, and the seed coat from the integuments of the ovule. • Seeds have been an important development in the reproduction and success of gymnosperm and angiosperm plants, relative to more primitive plants such as ferns, mosses and liverworts, which do not have seeds and use water-dependent means to propagate themselves. Seed plants now dominate biological niches on land, from forests to grasslands both in hot and cold climates. • The term "seed" also has a general meaning that antedates the above – anything that can be sown, e.g. "seed" potatoes, "seeds" of corn or sunflower "seeds". In the case of sunflower and corn "seeds", what is sown is the seed enclosed in a shell or husk, whereas the potato is a tuber. • In the angiosperms (flowering plants), the ovary ripens to a fruit which contains the seed and serves to disseminate it. Many structures commonly referred to as "seeds" are dry fruits. Sunflower seeds are sometimes sold commercially while still enclosed within the hard wall of the fruit, which must be split open to reach the seed. Different groups of plants have other modifications, the so-called stone fruits (such as the peach) have a hardened fruit layer (the endocarp) fused to and surrounding the actual seed. Nuts are the one-seeded, hard-shelled fruit of some plants with an indehiscent seed, such as an acorn or hazelnut.

Seed collection

Seeds may be collected from the field or bought from a market. Selection and collection of good seeds before sowing are directly related to the success of nursery business Therefore, the following points should be considered during the selection and collection of seeds: Select healthy and vigorous plants with desirable characters, i.e. plant shape, size, and fruit yield and quality etc. It is preferred to collect seeds from several plants in a particular area. Select mature, ripe, uniform and healthy fruits for seed extraction. Seeds should be extracted safely using proper method. Make the seeds free from pulp/juice by washing or cleaning. Select undamaged, healthy and viable seeds. Label the seeds when collected or describe the plant for later identification.

Advantages of Sexual Propagation

1) This is very simple and easy method of propagation. 2) Some species of trees, ornamental annuals and vegetables which cannot be propagated by asexual means should be propagated by this method. E.g. Papaya, Marigold, Tomato etc. 3) Hybrid seeds can be developed by this method. 4) New variety of crops are developed only by sexual method of propagation. 5) Root stocks for budding and grafting can be raised by this method. 6) The plants propagated by this method are long lived and are resistant to water stress. 7) Transmission of viruses can be prevented by sexual method. 8) Seed can be transported and stored for longer time for propagation.

Disadvantages of sexual propagation

1) Characteristics of seedling propagated by this method are not genetically true to type to that of their mother plant. 2) Plants propagated by sexual method requires long period for fruiting. 3) Plants grow very high, so they are difficult for intercultural practices like spraying, harvesting etc. 4) The plants which have no seeds cannot be propagated by this method. E.g. Banana, fig, Rose etc.

UNIT 4: TILLAGE PRACTICE

Tillage is the practice of cultivating the soil. After the land has been cleared and plant debris removed, it is often necessary to subject it to some form of tillage before the crop is planted. Tillage can be done using a variety of tools, including plows, harrows, and cultivators.

Purposes of Tillage

1. **Seed bed preparation:** Tillage loosens the soil and results in a seed bed suitable for seed germination and development of the young seedlings. A good seed bed should be moist and should not contain large lumps of soil that may prevent close contact between the planted seed and soil particle. It should not contain large quantities of un-decomposed organic matter.
2. **Control of weeds:** Often weeds growing on a fallow plot can be controlled by being ploughed under. Ploughing prior to cropping may also serve to kill the weeds present. Tillage between rows of growing crop can be an important method of weed control.
3. **Incorporation of organic matter into the soil:** Organic matter and crop residues can be incorporated into the soil through tillage. Once in the soil, they decomposed rapidly thus releasing the nutrients they contain to the growing plants. The incorporation of plant residues in the soil through tillage also serve to improved soil structure.
4. **Soil and water conservation:** Tillage often serves the purpose of breaking up the surface layers of the soil so that water is able to infiltrate more rapidly into the soil. This has the dual benefit of increasing the amount of water available in the soil for the crops and decreasing the amount of soil erosion caused by excessive run-off. For these reasons, land that is not to be cropped immediately may even be occasionally ploughed or tilled as a soil and water conservation measure.
5. **Improvement of the soil's physical condition:** The physical condition of the soil can be improved by tillage. For example, where the surface layers of the soil have formed a hardpan, it may be beneficial to break up the hardpan through tillage so that plants roots can penetrate more deeply and water could percolate into the lower layers of soil more easily.
6. **Increases soil aeration:** Tillage opens up the soil, thus improving soil aeration which increases the oxidation of chemical compounds in the soil and make them more soluble and available for the plants roots to absorb. Improved aeration is also beneficial to soil inhabiting organism.

Classification of tillage

There are two main types of tillage: primary tillage and secondary tillage.

- Primary tillage is deep tillage that breaks up the soil and prepares it for planting. It is typically done using a plow or other heavy-duty tillage tool.
- Secondary tillage is shallower tillage that is used to refine the soil and prepare it for planting. It is typically done using a harrow or cultivator.

Advantages of tillage

- Improved soil structure: Tillage can improve soil structure by breaking up large clods and aggregates. This improves drainage and aeration, which is essential for plant growth.
- Weed control: Tillage can help to control weeds by uprooting and burying them. This can reduce the need for herbicides.
- Improved seed-soil contact: Tillage can improve seed-soil contact by creating a smooth, even surface. This is important for germination and seedling establishment.

Disadvantages of tillage

- Soil erosion: Tillage can increase soil erosion by exposing the soil to wind and water.
- Loss of organic matter: Tillage can lead to the loss of organic matter from the soil. Organic matter is important for soil health and fertility.
- Increased fuel and labor costs: Tillage can increase fuel and labor costs.

Conservation tillage

Conservation tillage is a set of tillage practices that are designed to reduce soil erosion and improve soil health. Conservation tillage practices include:

- Minimum tillage: Minimum tillage involves disturbing the soil as little as possible. This can be done by using lighter tillage tools and reducing the number of tillage passes.
- No-till farming: No-till farming involves no tillage at all. Crops are planted directly into the residue of the previous crop.

Forms of tillage practice

A. **Ploughing:** This is one of the most ancient and most universal forms of tillage. A shear, which is pulled along by some power device, slices its way under the soil as it goes along, thereby loosening the soil and turning it over. The soil is left in lumps of various sizes and may require other operations in order to make a suitable seed bed. Implements used for Ploughing includes:
   i. Moldboard plough
   ii. Disc plough.

B. **Harrowing:** Planting may be done on the field after ploughing, but if the lumps of soil left after ploughing are too large, they must be broken up before planting. The conventional way of breaking up soil lumps is through harrowing. There are various kinds of animal-powered and tractor powered harrows but they all function by breaking up the lumps of soil and leaving an even soil surface.

C. **Mounding:** This involves the collection of the soil into more or less conical heaps or mounds. The mounds usually vary in height from 30-100cm, but are usually of approximately the same height on a particular farm. The distance from one mound to the next also varies but it is the distance that determines how much cropping can occur in the lower-lying spaces between the mounds. Mounding is the most common form of soil collection tillage in tropical Africa and is often associated with intercropping since it simultaneously permits two or more kinds of seed bed on the field; the top of the mounds is used for crops such as tubers which requires a deep layer of loose soil, the low-lying furrows are used for crops that have high requirement for water such as rice while the slopes of the mounds are used for intermediate crops.

Advantages of mounding

1. Providing a deep, loose seed bed which is particularly suitable for the development of roots and tubers.
2. Providing a variety of seed bed types on the same field, which may be advantageous to intercropping.
3. Elevating the seed bed and plant roots above the water table in fields with a high water table.
4. Mounding improves aeration for roots and facilitates the growth and development of underground tuber and root crops and the pods of groundnuts.
5. Mounding makes harvesting of tuber crops and pods of groundnuts easier.

Disadvantages of mounding

1. The major disadvantage of mounding is that it has not been mechanized and would probably be extremely difficult to mechanize.
2. Mounds impede the free movement of men and machinery through the field. For these reasons, mounding is mostly confined to traditional agriculture (with intercropping) to which it is well suited.

D. **Ridging:** This involves the collection of soil into elongated heaps called ridges. The distance between the ridges is variable, but is usually about one meter. Growing crops on ridges is quite common in tropical Africa. In the traditional settings, hoes and human labour are used to make ridges, but mechanical ridgers are also available and permit large areas to be ridged in a relatively short time. Ridging have the same advantages as mounding, but in addition ridging is an extremely useful measure for controlling erosion, particularly on sloping land. In such cases, ridging is done along the contour so that the flow of water down the slope is impeded. The water then flows along the furrows between the ridges. In order to further discourage the rapid flow of water within the furrows, cross ridges (also called cross bunds or tie ridges) are made across the furrows at intervals to connect one ridge to the next. Thus, ridging, when properly done, can decrease surface run-off, thereby reducing soil erosion and promoting water infiltration into the soil. Since it has been completely mechanized, ridging finds a place in modern agriculture, where it is the most common form of soil collection tillage.

E. **Bed making:** This is a form of land preparation which is more often discovered in horticultural and nursery practices than in field crop production. A bed is like a ridge in that it is elongated raised portion of the field. It is, however, usually much broader than a ridge, its top is flat and its length is usually not more than 20 m. Bed making is most commonly done with hand tools.

F. **Inter-tillage:** Inter-tillage is commonly practiced with respect to crops that are planted in rows. It involves tilling the areas between the crop rows. Its objectives are usually to control the weeds between the crop rows and to promote water percolation into the soil.

G. **Grading and terracing:** These are two land preparation operations which aim at effecting changes in the gross topography of the field. **Grading** is most commonly done where it is desired to use furrow or some other form of surface irrigation. The land is graded in such a way that the point of water supply is the highest on the field and the field slopes downwards from the water source. Grading of the crop field is also encouraged where the farmer intends to impound water for the flooding culture of rice or taro. If the land is sloping, it is graded to be relatively level. Elevated areas or dykes are constructed around the field so that the water does not flow away once impounded. **Terracing** is one of the methods of managing extremely sloping land for crop production. It creates a series of relatively flat horizontal portions alternating with vertical portions very similar to a flight of stairs. The flat portions are used for cropping. Terracing provides erosion control measures and permit cropping on land that would have otherwise remain useless for cropping.

UNIT 5: HAVE A BASIC UNDERSTANDING OF THE DEFINITION AND CLASSIFICATION OF WEEDS.

What is weed?

- Weeds are the plants, which grow where they are not wanted (Jethro Tull, 1731).
- In 1967 the Weed Science Society of America defined a weed as "a plant growing where it is not desired
- In 1989 the Society's definition was changed to define a weed as "any plant that is objectionable or interferes with the activities or welfare of man"
- A plant which contests with man for the possession of the soil" W.S. Blatchley 1912.

Common weeds around us.

Weed identification?

1. *Panicum maximum*
2. *Calopogonium mucunoides*
3. *Celosia iserti*
4. *Spigelia anthelmia*
5. *Centrosema pubescence*
6. *Aspillia africana*
7. *Ageratum conyzoides*

8. *Phyllanthus amarus*
9. *Euphorbia sp*
10. *Imperata cylindrical* etc

Effects of weeds on growing crops and man.

The effect of weeds on crops could either be **Harmful** or **Beneficial**

A. Weeds could be Harmful when they cause nuisance, disturb or obstruct with man activities

In what way then could weed be Harmful?

1. Reducing crop yield
2. Competition with crops
3. Reduce the value of properties
4. Serves as alternative host harbouring pest and diseases
5. Reduce the quality of agricultural produce
6. Interfere with other farm operations
7. Might cause delay in other farm operations
8. Cause health problems like hitching (Mucuna)
9. Allelopathic effect
10. Can cause fire hazard, especially during the dry season
11. Increase cost of production

B. Weed could be Beneficial to human

1. Weeds serves as vegetables for man eg *Talinum triangulare* (Waterleaf)
2. Weeds serves as fodder for animal eg (*Panicum maximum*)
3. Weeds serves as fuel
4. Weeds serves as manure for crops
5. Some weeds had medicinal values (*Cynodon dactylon* against Piles and Asthma)
6. Could serves as source of income

Weed Control Measures.

Weed control are measures taken to reduce the growth and spread of weeds

Weed control measures can be classified into five major categories

1. Preventive weed control: This refers to any measure taken to prevent weed from growing or emerging in a cultivated crops (cleaning of farm implement or equipment after use before moving to another location to use; make use of seeds free of weed seeds; screening of irrigation water before use, especially water from water bodies).

2. Mechanical Weed control: this involves the use farm equipment or machines in controlling weeds (eg tillage and use of weeding machines like brush cutter).
3. Cultural weed control: This refers to any measures taken to ensure that weed didn't grow or increase in number (eg. Crop rotation, avoid over grazing, good soil fertility).
4. Biological Weed control: this refers the use of weeds' natural enemies to control weed seeds germination and Spread of weed seeds. (eg. Cattle to control Panicum).
5. Chemical Weed Control: This refers to the use of chemical (Herbicides) in controlling weeds. The herbicide is either applied to soil or leaves (eg. Glyphoate, Paraquat, 2,4 D)

Know the different generic types of herbicides.

Generic Herbicides are herbicides produced and sold by a company other than the original manufacturer, but contained same active ingredient(s).

Glyphosate: Forceup, Tackle, Clearweed, Wurawura, Vinash,

Atrazine: Atraforce, Trilla

Paraquate: Paraforce, Raraq, Slasher

Some common herbicide in Nigeria and their purpose

Commonly used herbicides

1. Paraquat: this is contact herbicide used as at pre planting stage before land preparation to kill existing vegetation.
2. Diuron: this is a selective pre emergence herbicide used in root and tuber crops eg cassava, yam
3. Atrazine: this is a selective pre emergence herbicide used to control early weed infestation in crops like maize, cassava, yam
4. Glyphosate: this is systemic herbicide used as at pre planting stage before land preparation to kill existing vegetation.

UNIT 6: PRINCIPLES OF MANURING AND FERTILIZING

Manuring and fertilizing are essential agronomic practices used to maintain and improve soil fertility for sustainable crop production. These practices involve the application of organic materials (manures) and inorganic substances (fertilizers) to supply essential nutrients required for plant growth. The principles guiding their use are based on understanding soil-plant nutrient relationships, nutrient availability, crop requirements, and environmental conditions.

Soil fertility refers to the ability of soil to supply essential nutrients in adequate amounts and proper balance for plant growth. Over time, continuous cropping, erosion, leaching, and crop removal deplete soil nutrients, making external nutrient replenishment necessary. Manures and fertilizers serve as external sources of plant nutrients such as nitrogen, phosphorus, potassium, calcium, magnesium, and trace elements.

Manuring is the application of organic materials such as farmyard manure, compost, green manure, and plant residues to the soil. These organic materials not only supply nutrients but also improve soil structure, water-holding capacity, aeration, and microbial activity. They release nutrients slowly through decomposition, making them a long-term source of soil fertility improvement. The principle behind manuring is the enhancement of soil physical, chemical, and biological properties simultaneously.

Fertilizing, on the other hand, refers to the application of manufactured or processed nutrient sources that supply specific elements in concentrated and readily available forms. Fertilizers are designed to provide immediate plant nutrition and are usually classified into nitrogenous, phosphatic, potassic, and compound fertilizers. Unlike manures, fertilizers act quickly but do not significantly improve soil structure.

The first major principle of manuring and fertilizing is the principle of nutrient replacement. This principle states that nutrients removed from the soil through crop harvest must be replenished to prevent soil exhaustion. Every crop harvested removes a certain amount of nutrients, and failure to replace them leads to declining yields over time.

Another important principle is the principle of the limiting factor, which states that plant growth is determined by the nutrient that is in the shortest supply relative to the plant's needs. Even if all other nutrients are abundant, a deficiency in one essential element will limit crop performance. This principle emphasizes balanced fertilization rather than the application of a single nutrient.

The principle of balanced fertilization requires that nutrients be applied in correct proportions according to crop requirement and soil test results. Excess application of one nutrient may interfere with the uptake of another. For example, excessive nitrogen can reduce fruiting in some crops and increase susceptibility to pests and diseases.

The principle of timing and method of application is also critical. Nutrients must be applied at the time when crops need them most. For instance, phosphorus is often applied at planting because it supports root development, while nitrogen may be applied in split doses to match crop growth stages and reduce leaching losses. Proper placement methods such as banding, broadcasting, or side dressing influence nutrient efficiency.

Soil reaction and nutrient availability form another key principle. Soil pH affects the solubility and uptake of nutrients. In strongly acidic soils, nutrients like phosphorus become less available, while toxic elements such as aluminum may increase. Therefore, soil amendment practices such as liming may be necessary alongside fertilization to maintain optimal nutrient availability.

The principle of efficient nutrient use emphasizes minimizing nutrient losses through leaching, volatilization, erosion, and runoff. Organic manures help reduce such losses by improving soil structure and nutrient retention capacity, while proper fertilizer placement and timing reduce wastage.

Finally, the principle of integrated nutrient management combines both organic and inorganic sources of nutrients. This approach recognizes that neither manures nor fertilizers alone can sustain long-term soil fertility. The integration ensures immediate nutrient supply from fertilizers and long-term soil health improvement from organic inputs.

UNIT 7:UNDERSTANDING THE PRINCIPLES AND PRACTICES OF CROP PROTECTION

Crop protection is a scientific and practical approach aimed at safeguarding crops from organisms and factors that reduce their productivity, quality, and economic value. In tropical agricultural systems such as Nigeria, where environmental conditions favor rapid multiplication of pests and pathogens, effective crop protection is essential for sustainable food production.

Crop protection integrates knowledge from entomology, plant pathology, weed science, ecology, and agronomy to manage harmful organisms while maintaining environmental balance.

CONCEPT OF PESTS AND DISEASES IN CROP PRODUCTION

Definition of Pests

Pests are organisms that cause economic damage to crops either in the field or during storage. They are categorized as follows:

Pest Group	Examples	Type of Damage
Insect pests	Aphids, caterpillars, weevils	Chewing, sucking, boring
Vertebrate pests	Rats, birds, monkeys	Feeding, trampling
Disease pests	Fungi, bacteria, viruses, Nematodes	Plant infections
Weed pests	Spear grass, sedges	Competition
Storage pests	Grain weevils, moths, fungi	Post-harvest loss
Molluscs	Snails, slugs	Leaf and seedling damage

A pest becomes economically important when its population exceeds the **economic threshold level**, leading to measurable yield loss.

Definition of Diseases

Plant diseases are abnormal physiological conditions caused by pathogenic organisms such as fungi, bacteria, viruses, and nematodes. These pathogens disrupt normal plant functions like photosynthesis, respiration, and nutrient transport.

Key Concept: Pest–Host–Environment Interaction

Crop protection is based on the **disease triangle**, which involves:

- A susceptible host (crop)
- A virulent pest/pathogen
- A conducive environment

Disease or pest outbreak occurs only when all three factors interact favorably.

COMMON PESTS AND DISEASES IN FIELD AND STORAGE

Field Pests

- **Insect pests**: grasshoppers, aphids, caterpillars, stem borers
- **Vertebrates**: rodents, birds, monkeys

Field Diseases

- Fungal: leaf spot, rust, blight
- Bacterial: wilt, blight
- Viral: mosaic diseases
- Nematodes: root-knot nematodes (*Meloidogyne spp.*)

Storage Pests

- Insects: weevils (*Sitophilus spp.*), grain borers
- Rodents: rats and mice

Storage Diseases

- Fungal infections causing rot and mold (e.g., *Aspergillus*, *Penicillium*)
- Mycotoxin contamination (e.g., aflatoxins in grains)

IDENTIFICATION OF COMMON FIELD PESTS AND DISEASES

Identification of Pests

Pests are identified based on:

1. **Visual Observation**
- Presence of insects or animals
- Feeding marks (holes, cuts, tunnels)

2. **Damage Symptoms**
- Defoliation (leaf loss)
- Wilting or stunted growth
- Bored stems or fruits

Examples

- Stem borers in maize → holes in stems
- Aphids → sticky leaves (honeydew) and curling

Identification of Diseases

Diseases are identified through symptoms such as:

- Chlorosis (yellowing of leaves)
- Necrosis (dead tissues)
- Leaf spots and blights
- Wilting
- Mosaic patterns (viral diseases)

Examples

- Cassava mosaic disease → distorted leaves

- Rice blast → lesions on leaves
- Cocoa black pod → rotting pods

EFFECTS OF PESTS AND DISEASES ON CROPS

Effects on Field Crops

Pests and diseases affect crops in multiple ways:

1. Reduction in Yield

Damage to leaves, stems, roots, or reproductive parts reduces productivity.

2. Interference with Plant Physiology

- Reduced photosynthesis
- Poor nutrient uptake
- Impaired growth

3. Poor Crop Quality

- Discoloration
- Deformation
- Contamination

4. Increased Production Cost

More labor and inputs required for control.

Effects on Stored Products

- Weight loss due to feeding
- Contamination with droppings and insect parts
- Mold growth and toxin production
- Reduced market value and consumer acceptability

PEST CONTROL MEASURES (FIELD AND STORAGE)

Effective pest control requires **strategic selection and combination of methods**.

1. **Cultural controls**: Adjusting farming practices to prevent pest or disease buildup. These practices include;

   - Crop rotation

   - Plant spacing and pruning

   - Field sanitation (removing diseased plants or debris)

- Using resistant crop varieties

- Adjusting planting dates to avoid peak pest periods

Advantages

- Prevents pest buildup
- Environmentally safe
- Low cost

Disadvantages

- Requires planning and knowledge
- Might not be sufficient for severe infestations
- Can be labour intensive

2. **Biological controls**: Using natural enemies or biological agents to control pests. E.g.

- Using natural predators (e.g., ladybugs for aphids)

- Introducing beneficial insects (e.g., parasitoids)

- Using biopesticides (e.g., *Bacillus thuringiensis* for caterpillars)

- Encouraging biodiversity to support natural pest control

Advantages:

Environmentally friendly

Sustainable

Disadvantages:

Slow action and requires ecological balance

May not completely eliminate pests

3.Chemical controls: This involves the use of pesticides such as insecticides, fungicides, and rodenticides or other chemicals to control pests or diseases.

In the Field

- Spraying insecticides or fungicides

In Storage

- Dusting grains with insecticides
- Fumigation

Advantages:

- Quick and effective
- Suitable for severe infestations

Disadvantages:

- Environmental pollution
- Health risks to humans and animals
- Development of pest resistance

4. **Mechanical controls**: Using physical barriers or traps to control pests, such as

- Handpicking pests

- Using physical barriers (e.g., nets or row covers)

- Trapping pests (e.g., sticky traps or pitfall traps)

- Weeding by hand or with tools

- Heat or cold treatment in storage

Advantages

- Targets specific pests
- Less chemical use
- Eco-friendly

Disadvantages

- Labour intensive
- Might not work for large infestations
- Can be costly initially

5. Quarantine Methods:

- Restricting the movement of plants or plant materials to prevent the spread of pests and diseases

- Inspecting imported or exported plant materials

- Implementing biosecurity measures to prevent pest introduction or spread

6. Integrated pest management (IPM):

Integrated Pest Management (IPM) is a holistic approach to managing pests that combines physical, cultural, biological, and chemical controls to minimize harm to people, the environment, and beneficial organisms.

Example

- Crop rotation + resistant variety + limited pesticide use

IPM aims to:

1. **Identify and monitor pests**: Understand pest biology, behavior, and ecology.

2. **Set action thresholds**: Determine when pest populations reach levels that require action.

3. **Use a combination of controls**: Implement multiple management strategies, such as:

   - Cultural (e.g., crop rotation, sanitation)

   - Biological (e.g., natural predators, parasites)

   - Mechanical (e.g., handpicking, traps)

   - Chemical (e.g., targeted pesticides)

4. **Evaluate and adjust**: Continuously monitor and assess the effectiveness of IPM strategies and make adjustments as needed.

A**dvantages:**

Environmentally friendly

Reduces pesticide use

Sustainable long-term control

Disadvantages:

Requires technical knowledge and planning

May be costly initially

CONCLUSION

A thorough understanding of crop protection principles enables farmers to make informed decisions that minimize crop losses and maximize productivity. Sustainable crop protection depends on integrating multiple control strategies while maintaining ecological balance and ensuring food safety.

HARVESTING

Harvesting is the process of gathering or collecting mature or ripened or useful parts of plants. It can be done manually (by hand) or using tools like sickles, scythes, or machines like combine harvesters, depending on the scale and type of harvest. The collected crops are called “the harvest.” Harvesting marks the end of the growing season or cycle for a particular crop.

Factors Guiding Harvesting Time

1. Crop Maturity: This is the most critical factor guiding harvesting time because harvesting at the correct maturity stage ensures the best quality and yield. Different crops have different maturity indicators, and it is important to harvest when the crop is at its peak.

2. Environmental Conditions: Weather conditions like temperature, rainfall, and sunlight can affect crop growth and maturity, thereby influencing harvesting time.

3. Market Demands: Seasonal prices and consumer preferences can affect when crops are harvested.

4. Available Resources: The availability of labor, equipment, storage facilities, and transportation affects harvest efficiency and timing.

5. Crop Type and Variety: Different crops and varieties have different growth patterns and maturity periods, requiring specific harvesting techniques.

6. Intended Use of the Product: Whether the produce is for local consumption, export, or processing can determine harvesting time.

7. Harvest Season: Seasonal variations in weather and market demand influence harvesting time.

8. Disease and Pest Infestation: These can affect crop maturity and may force early or delayed harvesting.

Harvest Methods

Harvesting can be categorized into two main types:

1. Manual Harvesting: This involves the use of hand or tools in the harvesting of crops. Manual harvesting can come in the form of one or more of the following.

a. Hand Tools: Sickles, scythes, knives, etc., are used to cut and gather crops.

b. Picking and Gathering: Fruits and vegetables are usually picked by hand, especially on small farms and collected in small carts.

c. Manual Threshing: Techniques like beating stalks to release grains are common.

d. Digging: Root crops like potatoes, onions, cassava, and legumes like groundnut are harvested by pulling them out of the ground or using simple tools.

2. Mechanical Harvesting: This involve the of machines to harvest food commodities. Machines that can be used for this purpose include:

a. Combine Harvesters: These machines integrate reaping (cutting), threshing (separating grains), and cleaning in one process.

b. Reapers: These are machines that cut crops, often followed by threshing and cleaning separately.

c. Other Machines: These include specialized machines for certain crops e.g., fruit harvesters that shake trees or use air to loosen fruits.

TRANSPORTATION, STORAGE AND PRESERVATION OF HARVESTED CROPS

Proper management of harvested crops involves careful transportation, storage and preservation to minimize losses and maintain quality. Efficient transportation minimizes damage and delays. Proper storage protects crops from spoilage and pests. Preserving harvested crops extends their shelf life and ensures that they remain viable for longer periods.

Procedures for transporting harvested crops

1. Handle crops gently during loading and unloading to prevent bruising and damage.

2. Use refrigerated vehicles for perishable goods like fruits and vegetables to maintain optimal temperatures and extend shelf life.

3. Use appropriate packaging materials like corrugated boxes and crates to protect crops during transport.

4. Load vehicles carefully, distributing weight evenly and ensuring that packages are loaded in a way that facilitates easy unloading.

Procedures for storing harvested crops

1. Store crops in a cool, dry and well ventilated environment to prevent moisture build up and fungal growth.

2. Use appropriate storage structures like silos, barns or warehouses depending on the type and scale of crop.

3. Implement measures to protect stored crops from insects, rodents and other pests.

4. Maintain the moisture content of stored crops within the optimal range to prevent spoilage and ensure preservation.

Procedures for preserving harvested crops

1. Reduce moisture content through sun or other drying to inhibit microbial growth.

2. Lower temperatures using refrigerator or freezer to slow down enzyme activity and microbial growth.

3. Use heat processing to destroy microorganisms and create a hermetic seal to extend shelf life.

4. Use microorganisms to create desirable flavors and textures while also preserving food.

5. Use approved chemical preservatives with caution to inhibit spoilage.

6. Traditional techniques like storage in gourds, clay pots or covered pits which can be effective for certain crops and storage conditions can be used.

STORAGE STRUCTURES

Storage is the art of maintaining the quality of agricultural materials and preventing them from deterioration for a specific period of time, beyond their normal shelf life.

Crops grown for food fall into two broad categories: 1. Perishable crops 2. Non-perishable crops

Cereal grains can be stored for over a year and are considered non-perishable, whereas fruits and vegetables are perishable crops because they deteriorate within a few days after harvesting.

Storage Structures for Crops

Structure	Crop
1. Rhombus	Grains such as guinea corn and millet that have not been threshed
2. Cribs	
3. Barns	Fresh yams, maize, guinea corn
4. Bags	Grains like rice and beans
5. Drums/ bins	Small grains
6. Silo	Bulk storage of shelled grains on a large scale for a long period

PRIMARY PROCESSING OF CROPS

Primary processing is the first level of processing or the initial steps taken after harvest to prepare agricultural products for consumption or further processing.

Processes Involved in Primary Processing

1. Cleaning: This involves the removal of dirt and other foreign materials from harvested crops.
2. Sorting: This is the separation of crops based on size, shape, colour, or other characteristics.
3. Grading: This involves classifying crops into different quality levels based on specific criteria such as size, colour, and shape.
4. Hulling/Shelling: This is the removal of outer layers or shells of grains, nuts, and seeds.
5. Milling/Grinding: This the process of converting grains into flour, meal, or other forms.
6. Drying: This is the removal of moisture from crops to prevent spoilage and improve storage life.
7. Pounding: This involves the use of pestle and mortar or mechanical means to break down or process crops.
8. Parboiling: This refers to the partial cooking of grains in water to improve cooking and milling properties.
9. Tempering/Soaking: This involves preparing grains for further processing by softening them with water.
10. Freezing: This is the process of preserving food by lowering its temperature, thereby slowing down or stopping microbial and enzymatic activities.